Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/24/2006
PI Name:	Halas, Naomi Jean Ph.D.		
Project Title:	Metal Nanoshell Functionalization and Materials Assembly: Effects of Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	halas@ece.rice.edu	**Fax:**	FY 713-348-5686
PI Organization Type:	UNIVERSITY	**Phone:**	713-348-5611
Organization Name:	Rice University		
PI Address 1:	ECE Department Mail Stop 366		
PI Address 2:	P.O. Box 1892, 6100 South Main		
PI Web Page:	http://www.ece.rice.edu/		
City:	Houston	**State:**	TX
Zip Code:	77005	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	02/01/1998	**End Date:**	01/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Cobb, Sharon	**Contact Phone:**	256 544-7791
Contact Email:	sharon.cobb@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, T. Randall (University of Houston)		
Grant/Contract No.:	NAG8-1467		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		
Task Description:	We are investigating the limits of growth under terrestrial gravitational conditions on the core sphericity and the shell homogeneity, and the size dispersion, as a function of growth parameters. We are advancing and refining our synthetic method for capping dielectric and semiconductor nanoparticles with metallic shells, and we are developing self-assembly methods for condensing metal nanoshell particles into nanocomposite materials. Our task is to understand the potential role of reduced gravity conditions on the optimized growth of metal nanoshell (dielectric core/metallic shell) particles. This composite nanoparticle has highly unique optical properties: it has an absorption and/or scattering resonance frequency that can be adjusted by varying the diameter of the core and the thickness of the outer metallic shell. This		

	resonance frequency can be adjusted across hundreds of nanometers in the visible and infrared regions of the spectrum. This property gives rise to a wide range of applications for these particles. If microgravity conditions can improve the growth of these particles, it will lead to an improvement of their optical properties and tuning range relative to terrestrially-grown particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)